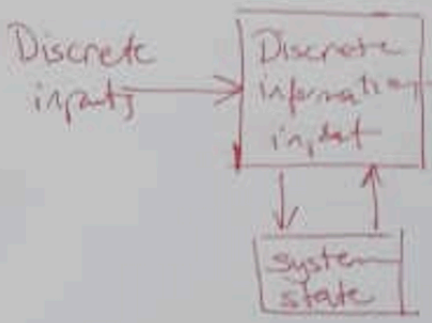
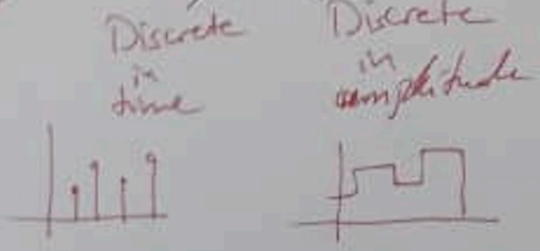
* Digital System:

W(1).2

(1)



Discrete: Non continuous



Types of digital systems:

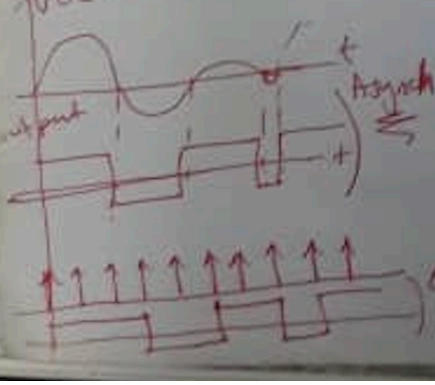
No state present
- Combinational logic system:
The output(s) are only depend on current input(s)
output(s) = f(input(s))

State present
- The systems with memory. Hence the output is depends on inputs and the previous state(s).
~~output(s) = f(inputs, prev state, current)~~
outputs = f(inputs, states)

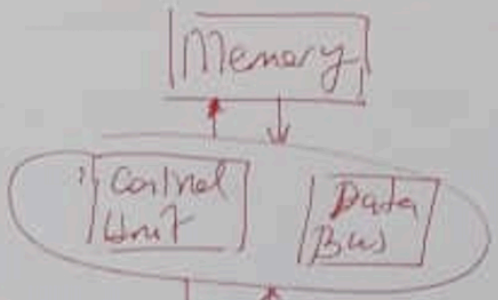
Synchronous sys.
* Clocking signal exists and current output state is only set/updated by clocking signal

Asynchronous sys.
* No clocking signal. The outputs are changing as soon as the current inputs change.

The system detects if the signal >0 or not



Digital Compute Example.

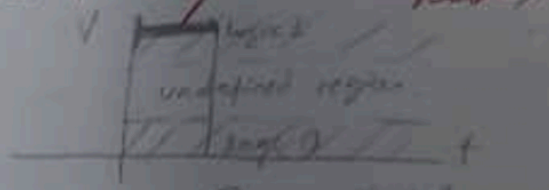


Inputs: keyboard, mouse, sensors, v.6.

outs.: Monitor, speakers, leds, etc.

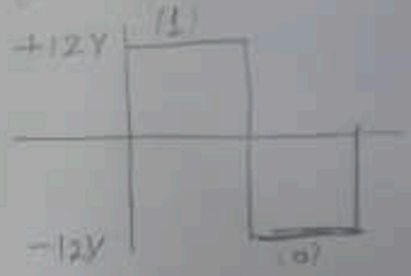
* Information Representation:
  → Information variables are represented by physical quantities. (Ex: Temperature)
  → for Digital systems variables take on discrete values.
  → Two level/binary values are the most prevalent values in digital systems.
    → 1 or 0 (Digits)
    → high/low
    → true/false
    → on or off

Why two values? It is robust! (less sensitive to noise)



Binary digital signal

the voltage difference between logic and logic 0 defines robustness to noise or ability.



→ Since we have only 1 or 0 in a digit we will use binary number system in digital system design.

NUMBER SYSTEMS

* DECIMAL NUMBER SYSTEM:

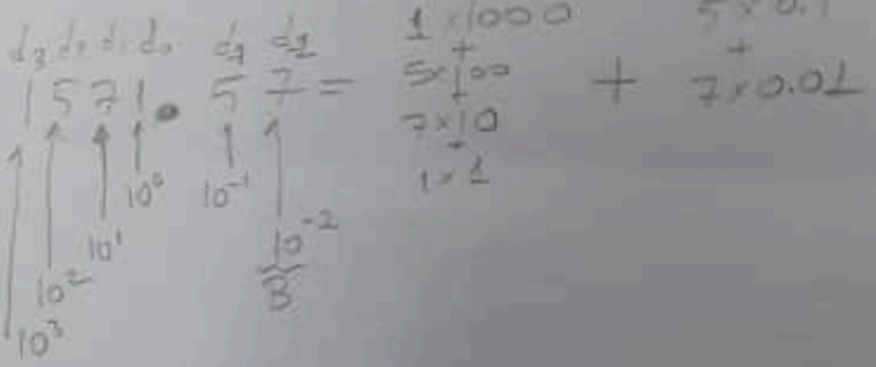
→ The number system that we are using in everyday life.

→ The base is 10. (That means every digit may take 10 different number values) [0 - 9]

→ Why we use it? Maybe 10 fingers.

→ A positional number system. The digit weight depends on its position. Non-positional number systems: Roman Numbers

→ Number has an integer part and a fractional part.



Here Base, (B = 10)

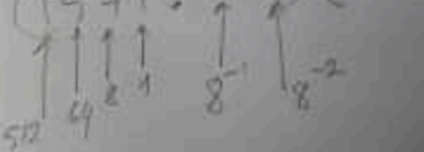
The number: $d_3 B^3 + d_2 B^2 + d_1 B^1 + d_0 B^0 + d_{-1} B^{-1} + d_{-2} B^{-2}$

This formula converts any base number to 10 base.

* Octal NUMBER SYSTEMS

The same rules, B = 8.

$(1571.74)_8 = (889.9375)_{10}$



Each digit can only take the values 0 to 7

* BINARY NUMBER SYSTEM

→ Each digit can only be 0 or 1.

→ BInary digITS called bits.

$(101.01)_2$
$2^2\ 2^1\ 2^0\ \ 2^{-1}\ 2^{-2} \longrightarrow (5.25)_{10}$

→ 4 bits group : A Nibble.

8 bits group : A byte.

* HEXA DECIMAL NUMBER SYSTEM

→ Base 16. (0, 1, 2, ..., 9, (10) A, B, C, D, E, F) 15

$(1E5.7A)_{16}$
$16^2\ 16^1\ 16^0\ \ 16^{-1}\ 16^{-2} \longrightarrow (485.476\,5625)_{10}$

DECIMAL (INTEGER) to BINARY or OCTAL or HEX. CONVERSION

* Divide the number by the required base (2, 8, 16)
* Take the remainder as a coefficient
* Take the quotient and repeat the divisor until the quotient is smaller than base.

Example $(113)_{10}$

HEX:
$\begin{array}{r|l} 113 & 16 \\ 112 & 7 \\ \hline 1 & \end{array}$ $(71)_{16}$

OCT
$\begin{array}{r|l} 113 & 8 \\ 112 & 14 \\ \hline 1 & 6 \end{array}$ $(161)_8$

BIN
$\begin{array}{r|l} 113 & 2 \\ \hline 1 & 56 \\ & 0 \end{array}$ $\begin{array}{r|l} 56 & 2 \\ \hline 0 & 28 \\ & 0 \end{array}$ $\begin{array}{r|l} 28 & 2 \\ \hline 0 & 14 \\ & 0 \end{array}$ $\begin{array}{r|l} 14 & 2 \\ \hline 0 & 7 \\ & 1 \end{array}$ $\begin{array}{r|l} 7 & 2 \\ \hline 1 & 3 \\ & 1 \end{array}$ $\begin{array}{r|l} 3 & 2 \\ \hline 1 & 1 \end{array}$

$(1110001)_2$

DECIMAL (Fraction) to OCT, HEX, BIN. CONVERS.

* Multiply the fraction by the base.
* Take the integer part as a coefficient
* Take the resultant fraction and repeat the operation until the resultant fraction is zero

EX.: $(0.625)_{10}$

HEX

a_{-1} 0.625 × 16 = 10.0

$(0.A)_{16}$

OCT

a_{-1} 0.625 × 8 = 5.0

$(0.5)_8$

BIN

a_{-1} 0.625 × 2 = 1.25

a_{-2} 0.25 × 2 = 0.5

a_{-3} 0.5 × 2 = 1.0

$(0.101)_2$

MOST SIGNIFICANT & LESS SIG. BITS.
(MSB) (LSB)

→ For integers the most significant bit is at the left most. LSB is right most.

→ For fractions, MSB → close to the fraction point. (Simply left most again); LSB → right most

BINARY to OCTAL OR HEX